

Agostino Scalzullo

Audio Software & DSP Engineer

C++ · DSP · Max/MSP · JUCE · VST3/AU · Real-Time Audio · Hardware Integration · MIDI/MPE

London, UK · OSTIN SOLO · VSTOPIA
contact@ostinsolo.co.uk · Live Portfolio · GitHub

PROFILE

Audio software and DSP engineer building real-time music systems across native C++, Max/MSP, JUCE plug-ins and interactive hardware. Developer of commercial OSTIN SOLO music software and founder of VSTOPIA, with hands-on experience taking systems from musical concept and rapid prototyping through native implementation, hardware integration, testing, packaging and release. Work spans DSP, MIDI/MPE, networked audio and selected machine-learning audio systems.

CORE SKILLS

Audio / DSP: C/C++, Objective-C++, JUCE, Max SDK / Max/MSP, VST3/AU, real-time DSP, MIDI/MPE, synthesis and modulation; architectural familiarity with RNBO and its C++ / WebAssembly export pipeline

Hardware / embedded: Arduino, ESP32, MCP23017 I/O expansion, firmware integration and modification, physical controls and sensors, microcontroller-to-host integration

Real-time / platform: WebRTC, Opus, RTP MIDI, WebSockets, AVFoundation, platform APIs, Swift/SwiftUI, macOS/iOS/Windows

Engineering / deployment: profiling, benchmark design, parity and regression testing, cross-platform builds, conditional compilation, model/runtime packaging, Developer ID signing and notarization

Audio ML / vision: source separation, BS-RoFormer, Demucs, MossFormer2, ECAPA, MLX, Core ML, CLAP embeddings, Depth Anything, MediaPipe, ARKit

SELECTED WORK

DSm - Dynamic Split Module

Neural separation, restoration and web-sampling environment for Ableton Live, with 63 model workflows spanning Demucs, RoFormer and related architectures. Covered by Sound On Sound and listed in the Cycling '74 project ecosystem.

[Product](#) · [Sound On Sound](#) · [Cycling '74](#)

Native RoFormer External & VST3/AU - Work in Progress

Native C++/MLX implementation bringing six-stem BS-RoFormer separation into a Max/MSP external and an in-development plug-in architecture. Separate Apple Silicon benchmarking of the MLX implementation against the python-audio-separator MPS baseline showed a **median 1.85x speedup**, reaching **2.5x on MelBand RoFormer**. The native C++/MLX six-stem runtime achieved **RTF ~ 0.36 (~2.8x real time)** on M4 Pro with bit-exact parity to the Python MLX reference; C++ and Python MLX performance were essentially matched in the fixed benchmark.

AUDIOENCE

JUCE/C++ networked collaboration plug-in architecture combining real-time audio, MIDI, WebRTC/P2P communication, Opus/PCM modes, resampling and DAW synchronisation across macOS and Windows.

[Official project](#) · [Technical manual](#)

Native Max / MPE / Hardware Systems

Native C++ Max externals and hardware-integration work spanning multitouch, MPE, computer sensors, Arduino/ESP32 microcontrollers, MCP23017 I/O expansion, Depth Anything and AprilTag workflows. Includes Objective-C++ bridge layers, platform APIs, firmware integration/modification, native model/runtime integration and ongoing JUCE VST3/AU migration. Earlier physical-computing prototypes included an MCP23017 encoder/button controller, 12-channel capacitive-touch MIDI control, ribbon-sensor pitch control, and phone/accelerometer-controlled servo positioning, first over an external Bluetooth module and later ESP32.

[Public product catalogue](#)

SELECTED WORK - CONTINUED

SoundCloner - Private R&D

Native synthesis and audio-analysis research platform combining deterministic DSP, oscillator/envelope/modulation systems, parameter/genome representations, similarity scoring, candidate retrieval, ML inference and large synthetic evaluation infrastructure.

Real-Time Target-Speaker Voice Isolation - Private R&D

Real-time target-speaker extraction using MossFormer2 + ECAPA, with speaker enrollment and suppression of competing voices.

Sample-Space Exploration - Private R&D

CLAP/audio embeddings, classifier training, similarity retrieval and Three.js spatial exploration of audio corpora, supported by benchmark and evaluation tooling.

iOS Computer Vision - Private R&D

Swift/SwiftUI iOS application work using ARKit, SceneKit, AVFoundation and MediaPipe Tasks Vision for real-time facial/eye landmark processing and guided physical-makeup workflows.

ENGINEERING VALIDATION

Repeatable benchmarking and regression methodology across projects: controlled baselines, latency/throughput profiling, held-out ML evaluation, parity testing, hardware-in-the-loop checks and cross-machine validation across Apple Silicon, Intel Mac and Windows execution paths where supported.

EXPERIENCE

Founder & Audio Software Developer - VSTOPIA | 2023-Present

Built an independent software ecosystem spanning product development, web/backend infrastructure, distribution, licensing, updates and developer workflows.

Independent Audio Software Developer - OSTIN SOLO | 2018-Present

Commercial and experimental audio technology across DSP, Max for Live, native Max externals, plug-ins, machine learning, computer vision, MPE/hardware control and networked music systems.

Founder, Artistic Director & Experience Designer - Macrowave / Cacao Discoteca / Basement | 2014-2016

Directed electronic-music and multimedia programmes in Italy for audiences up to 3,000, spanning artist curation, production, projection mapping, scenography and interactive installations.

EDUCATION

Music Production - SAE Institute | 2019-2020 · Recording, editing, mixing, mastering, sound design and synthesis

Electronic Music & Musicianship - Italian Conservatory | c. 2004-2009

RECOGNITION & PUBLIC EVIDENCE

Sound On Sound coverage of DS_m · Cycling '74 official project listing · OSTIN SOLO commercial product catalogue · public AUDIOENCE platform and technical documentation.

Full technical portfolio: ostinsolo.github.io/ostin-solo-portfolio/